Healthcare Disease Risk Prediction System
Corporate Technical Documentation

This document provides a professional technical walkthrough of the Healthcare Disease Risk Prediction notebook. It explains the purpose, business value, machine learning rationale, and implementation intent behind each code section. The documentation is designed for portfolio presentation, technical interviews, project reviews, and stakeholder communication.

Cell 1: Technical Explanation

Code Overview

```
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from pathlib import Path

from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.impute import SimpleImputer
from sklearn.model_selection import train_test_split, StratifiedKFold, GridSearchCV,
cross_val_score
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import (
accuracy_score, precision_score, recall_score, f1_score,
roc_auc_score, classification_report, confusion_matrix, roc_curve
)
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier,
ExtraTreesClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC

try:
from xgboost import XGBClassifier
except Exception:
XGBClassifier = None

sns.set_theme(style="whitegrid", context="talk")
plt.rcParams["figure.figsize"] = (10, 6)
pd.set_option("display.max_columns", None)
pd.set_option("display.width", 120)
```

Purpose: This cell is part of the end-to-end healthcare machine learning workflow. It contributes to data ingestion, validation, preprocessing, feature engineering, model development, evaluation, visualization, or deployment activities depending on its position in the notebook.

Why This Code Exists: Every component is included to improve model reliability, reproducibility, maintainability, and business usefulness. The workflow follows common enterprise data science practices including data quality checks, missing value handling, pipeline construction, cross-validation, model benchmarking, hyperparameter tuning, and artifact generation.

Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.

Cell 2: Technical Explanation

Code Overview

```
df = pd.read_csv("/content/health.csv.csv")
df.columns = [c.strip().lower().replace(" ", "_") for c in df.columns]

display(df.head())
print("Shape:", df.shape)
```

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Cell 3: Technical Explanation

Code Overview

```
display(df.info())
display(df.describe(include="all").T)
```

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Cell 4: Technical Explanation

Code Overview

```
missing = pd.DataFrame({
"missing_count": df.isna().sum(),
"missing_percent": (df.isna().mean() * 100).round(2)
}).sort_values("missing_percent", ascending=False)

display(missing)
```

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Cell 5: Technical Explanation

Code Overview

```
print("Duplicate rows before:", df.duplicated().sum())
df = df.drop_duplicates().reset_index(drop=True)
print("Shape after duplicate removal:", df.shape)
```

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Cell 6: Technical Explanation

Code Overview

```
target = "disease_risk"
id_col = "patient_id" if "patient_id" in df.columns else None

num_cols = ["age", "bmi", "blood_pressure", "cholesterol", "glucose"]
cat_cols = ["gender", "region"]
```

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Cell 7: Technical Explanation

Code Overview

```
for c in num_cols:
df[c] = pd.to_numeric(df[c], errors="coerce")
```

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Cell 8: Technical Explanation

Code Overview

```
df["bp_glucose_ratio"] = df["blood_pressure"] / (df["glucose"] + 1e-5)
df["cholesterol_bmi_ratio"] = df["cholesterol"] / (df["bmi"] + 1e-5)
df["age_bmi_interaction"] = df["age"] * df["bmi"]
df["risk_score_raw"] = (
0.25 * df["age"] +
0.25 * df["bmi"] +
0.25 * df["blood_pressure"] +
0.125 * df["cholesterol"] +
0.125 * df["glucose"]
)

df["age_group"] = pd.cut(
df["age"],
bins=[0, 30, 45, 60, 100],
labels=["Young", "Adult", "Middle_Aged", "Senior"]
)

df["bmi_group"] = pd.cut(
df["bmi"],
bins=[0, 18.5, 25, 30, 100],
labels=["Underweight", "Normal", "Overweight", "Obese"]
)
```

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Cell 9: Technical Explanation

Code Overview

```
for c in ["age", "bmi", "cholesterol", "glucose"]:
df[f"{c}_missing"] = df[c].isna().astype(int)
```

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Cell 10: Technical Explanation

Code Overview

```
num_imputer = SimpleImputer(strategy="median")
cat_imputer = SimpleImputer(strategy="most_frequent")

df[num_cols] = num_imputer.fit_transform(df[num_cols])
df[cat_cols] = cat_imputer.fit_transform(df[cat_cols])

df["age_group"] = df["age_group"].astype(str)
df["bmi_group"] = df["bmi_group"].astype(str)
```

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Cell 11: Technical Explanation

Code Overview

```
clip_cols = num_cols + ["bp_glucose_ratio", "cholesterol_bmi_ratio", "age_bmi_interaction",
"risk_score_raw"]

for c in clip_cols:
q1, q3 = df[c].quantile([0.25, 0.75])
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
df[c] = df[c].clip(lower, upper)
```

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Cell 12: Technical Explanation

Code Overview

```
corr = df.select_dtypes(include=np.number).corr()
corr = corr.rename(columns={
"blood_pressure": "BP",
"cholesterol": "Chol",
"disease_risk": "Risk"
}, index={
"blood_pressure": "BP",
"cholesterol": "Chol",
"disease_risk": "Risk"
})
```

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Cell 13: Technical Explanation

Code Overview

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid", context="talk")
plt.rcParams.update({
"figure.facecolor": "white",
"axes.facecolor": "#FCFCFC",
"axes.edgecolor": "#D0D0D0",
"axes.labelsize": 13,
"axes.titlesize": 18,
"xtick.labelsize": 11,
"ytick.labelsize": 11,
"font.family": "DejaVu Sans"
})

fig, ax = plt.subplots(
1, 2,
figsize=(18, 6.5),
gridspec_kw={"width_ratios": [1, 1.35]},
constrained_layout=True
)

# Left: Target distribution
target_palette = ["#4C78A8", "#F58518"]
sns.countplot(
data=df,
x=target,
ax=ax[0],
palette=target_palette,
edgecolor="white",
linewidth=1.5,
saturation=0.95
)

ax[0].set_title("Target Distribution", fontsize=18, fontweight="bold", pad=14)
ax[0].set_xlabel("")
ax[0].set_ylabel("Count", fontsize=13, labelpad=10)
ax[0].grid(axis="y", linestyle="--", alpha=0.25)
ax[0].set_facecolor("#FCFCFC")
ax[0].spines["top"].set_visible(False)
ax[0].spines["right"].set_visible(False)
ax[0].spines["left"].set_color("#D0D0D0")
ax[0].spines["bottom"].set_color("#D0D0D0")
ax[0].tick_params(axis="both", labelsize=12)

for p in ax[0].patches:
h = p.get_height()
ax[0].annotate(
f"{int(h)}",
(p.get_x() + p.get_width() / 2., h),
ha="center",
```

```
va="bottom",
fontsize=12,
fontweight="bold",
xytext=(0, 6),
textcoords="offset points"
)

# Right: Correlation heatmap
corr = df.select_dtypes(include=np.number).corr()

mask = np.triu(np.ones_like(corr, dtype=bool))

sns.heatmap(
corr,
mask=mask,
ax=ax[1],
cmap=sns.diverging_palette(230, 20, as_cmap=True),
vmin=-1,
vmax=1,
center=0,
annot=True,
fmt=".2f",
annot_kws={"size": 9, "weight": "bold"},
square=True,
linewidths=0.8,
linecolor="white",
cbar_kws={"shrink": 0.85, "label": "Correlation", "pad": 0.02}
)

ax[1].set_title("Numeric Correlation", fontsize=18, fontweight="bold", pad=14)
ax[1].set_xlabel("")
ax[1].set_ylabel("")

ax[1].tick_params(axis="x", labelrotation=45, labelsize=10, pad=2)
ax[1].tick_params(axis="y", labelrotation=0, labelsize=10)

for label in ax[1].get_xticklabels():
label.set_horizontalalignment("right")
label.set_rotation_mode("anchor")

for spine in ax[1].spines.values():
spine.set_visible(False)

fig.suptitle("Exploratory Overview", fontsize=22, fontweight="bold", y=1.03)
plt.show()
```

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Cell 14: Technical Explanation

Code Overview

```
sns.set_theme(style="whitegrid", context="talk")
plt.rcParams.update({
"figure.facecolor": "white",
"axes.facecolor": "#FCFCFC",
"axes.edgecolor": "#D0D0D0",
"axes.labelcolor": "#1f2937",
"text.color": "#1f2937",
"xtick.color": "#374151",
"ytick.color": "#374151",
"font.family": "DejaVu Sans"
})

features = ["age", "bmi", "blood_pressure", "cholesterol"]
titles = ["Age", "BMI", "Blood Pressure", "Cholesterol"]
colors = ["#4C78A8", "#F58518", "#54A24B", "#E45756"]

fig, axes = plt.subplots(2, 2, figsize=(16, 10), constrained_layout=True)
fig.suptitle(
"Clinical Feature Distributions",
fontsize=22,
fontweight="bold",
y=1.02
)

for ax, col, title, color in zip(axes.flat, features, titles, colors):
sns.histplot(
data=df,
x=col,
kde=True,
bins=24,
stat="density",
color=color,
edgecolor="white",
linewidth=1.0,
alpha=0.92,
ax=ax
)

ax.set_title(title, fontsize=17, fontweight="bold", pad=10)
ax.set_xlabel("")
ax.set_ylabel("Density")
ax.grid(axis="y", linestyle="--", alpha=0.25)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.spines["left"].set_color("#D0D0D0")
ax.spines["bottom"].set_color("#D0D0D0")
ax.tick_params(labelsize=11)
ax.margins(x=0.02)

fig.suptitle("Clinical Feature Distributions", fontsize=22, fontweight="bold", y=1.02)

plt.show()
```

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Cell 15: Technical Explanation

Code Overview

```
drop_cols = [target] + ([id_col] if id_col else [])
X = df.drop(columns=drop_cols)
y = df[target]
```

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Cell 16: Technical Explanation

Code Overview

```
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y
)
```

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Cell 17: Technical Explanation

Code Overview

```
all_num_cols = [
"age", "bmi", "blood_pressure", "cholesterol", "glucose",
"bp_glucose_ratio", "cholesterol_bmi_ratio", "age_bmi_interaction",
"risk_score_raw", "age_missing", "bmi_missing", "cholesterol_missing", "glucose_missing"
]

all_cat_cols = ["gender", "region", "age_group", "bmi_group"]
```

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Cell 18: Technical Explanation

Code Overview

```
numeric_transformer = Pipeline(steps=[
("imputer", SimpleImputer(strategy="median")),
("scaler", StandardScaler())
])

categorical_transformer = Pipeline(steps=[
("imputer", SimpleImputer(strategy="most_frequent")),
("encoder", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(transformers=[
("num", numeric_transformer, all_num_cols),
("cat", categorical_transformer, all_cat_cols)
])
```

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Cell 19: Technical Explanation

Code Overview

```
models = {
"Logistic Regression": LogisticRegression(max_iter=2000, class_weight="balanced"),
"Decision Tree": DecisionTreeClassifier(random_state=42),
"Random Forest": RandomForestClassifier(random_state=42, class_weight="balanced"),
"Extra Trees": ExtraTreesClassifier(random_state=42, class_weight="balanced"),
"Gradient Boosting": GradientBoostingClassifier(random_state=42),
"KNN": KNeighborsClassifier()
}

if XGBClassifier is not None:
models["XGBoost"] = XGBClassifier(
eval_metric="logloss",
random_state=42,
n_jobs=-1
)
```

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Cell 20: Technical Explanation

Code Overview

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
bench_results = []

for name, model in models.items():
pipe = Pipeline([
("preprocess", preprocessor),
("model", model)
])
scores = cross_val_score(pipe, X_train, y_train, cv=cv, scoring="accuracy")
bench_results.append({
"Model": name,
"CV Mean Accuracy": scores.mean(),
"CV Std": scores.std()
})

bench_df = pd.DataFrame(bench_results).sort_values("CV Mean Accuracy", ascending=False)
display(bench_df)
```

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Cell 21: Technical Explanation

Code Overview

```
param_grids = {
"Logistic Regression": {
"model__C": [0.1, 1, 10],
"model__solver": ["liblinear", "lbfgs"]
},
"Decision Tree": {
"model__max_depth": [3, 5, 7, None],
"model__min_samples_split": [2, 5, 10]
},
"Random Forest": {
"model__n_estimators": [200, 300, 500],
"model__max_depth": [None, 5, 10],
"model__min_samples_split": [2, 5]
},
"Extra Trees": {
"model__n_estimators": [200, 300],
"model__max_depth": [None, 5, 10]
},
"Gradient Boosting": {
"model__n_estimators": [100, 200],
"model__learning_rate": [0.01, 0.05, 0.1],
"model__max_depth": [2, 3, 4]
},
"KNN": {
"model__n_neighbors": [3, 5, 7, 9],
"model__weights": ["uniform", "distance"]
}
}

if XGBClassifier is not None:
param_grids["XGBoost"] = {
"model__n_estimators": [100, 200, 300],
"model__max_depth": [3, 4, 5],
"model__learning_rate": [0.01, 0.05, 0.1],
"model__subsample": [0.8, 0.9],
"model__colsample_bytree": [0.8, 0.9]
}
```

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Cell 22: Technical Explanation

Code Overview

```
best_models = []
best_score = -1
best_name = None
best_pipe = None

for name, model in models.items():
pipe = Pipeline([
("preprocess", preprocessor),
("model", model)
])

grid = GridSearchCV(
pipe,
param_grids[name],
cv=cv,
scoring="accuracy",
n_jobs=-1
)
grid.fit(X_train, y_train)

best_models.append({
"Model": name,
"Best CV Accuracy": grid.best_score_,
"Best Params": grid.best_params_
})

if grid.best_score_ > best_score:
best_score = grid.best_score_
best_name = name
best_pipe = grid.best_estimator_

tuning_df = pd.DataFrame(best_models).sort_values("Best CV Accuracy", ascending=False)
display(tuning_df)
print("Best tuned model:", best_name)
```

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Cell 23: Technical Explanation

Code Overview

```
y_pred = best_pipe.predict(X_test)
y_prob = best_pipe.predict_proba(X_test)[:, 1]

final_metrics = {
"Accuracy": accuracy_score(y_test, y_pred),
"Precision": precision_score(y_test, y_pred),
"Recall": recall_score(y_test, y_pred),
"F1 Score": f1_score(y_test, y_pred),
"ROC AUC": roc_auc_score(y_test, y_prob)
}

final_metrics_df = pd.DataFrame([final_metrics], index=[best_name])
display(final_metrics_df)
print(classification_report(y_test, y_pred))
```

Purpose: This cell is part of the end-to-end healthcare machine learning workflow. It contributes to data ingestion, validation, preprocessing, feature engineering, model development, evaluation, visualization, or deployment activities depending on its position in the notebook.

Why This Code Exists: Every component is included to improve model reliability, reproducibility, maintainability, and business usefulness. The workflow follows common enterprise data science practices including data quality checks, missing value handling, pipeline construction, cross-validation, model benchmarking, hyperparameter tuning, and artifact generation.

Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.

Cell 24: Technical Explanation

Code Overview

```
sns.set_theme(style="white", context="talk")
plt.rcParams.update({
"figure.facecolor": "white",
"axes.facecolor": "#FCFCFC",
"axes.edgecolor": "#D0D0D0",
"axes.labelcolor": "#1f2937",
"text.color": "#1f2937",
"xtick.color": "#374151",
"ytick.color": "#374151",
"font.family": "DejaVu Sans"
})

cm = confusion_matrix(y_test, y_pred)
labels = ["No Risk", "Risk"]

fig, ax = plt.subplots(figsize=(7.5, 6.5), constrained_layout=True)

sns.heatmap(
cm,
annot=True,
fmt="d",
cmap=sns.light_palette("#2563EB", as_cmap=True),
cbar=True,
square=True,
linewidths=1.2,
linecolor="white",
annot_kws={"size": 14, "weight": "bold"},
ax=ax
)

ax.set_title(
f"Confusion Matrix - {best_name}",
fontsize=20,
fontweight="bold",
pad=16
)
ax.set_xlabel("Predicted Class", fontsize=13, labelpad=10)
ax.set_ylabel("Actual Class", fontsize=13, labelpad=10)
ax.set_xticklabels(labels, rotation=0, fontsize=12)
ax.set_yticklabels(labels, rotation=0, fontsize=12)

ax.tick_params(length=0)
for spine in ax.spines.values():
spine.set_visible(False)

fig.text(
0.5, 0.98,
"Model performance on hold-out test set",
ha="center",
fontsize=12,
color="#6B7280"
)
```

```
plt.show()
```

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Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.

Cell 25: Technical Explanation

Code Overview

```
sns.set_theme(style="white", context="talk")
plt.rcParams.update({
"figure.facecolor": "white",
"axes.facecolor": "#FCFCFC",
"axes.edgecolor": "#D0D0D0",
"axes.labelcolor": "#1f2937",
"text.color": "#1f2937",
"xtick.color": "#374151",
"ytick.color": "#374151",
"font.family": "DejaVu Sans"
})

fpr, tpr, _ = roc_curve(y_test, y_prob)
auc_score = roc_auc_score(y_test, y_prob)

fig, ax = plt.subplots(figsize=(7.5, 6.5), constrained_layout=True)

ax.plot(
fpr, tpr,
color="#2563EB",
linewidth=3,
label=f"{best_name} (AUC = {auc_score:.3f})"
)
ax.fill_between(fpr, tpr, color="#2563EB", alpha=0.12)
ax.plot(
[0, 1], [0, 1],
linestyle="--",
color="#9CA3AF",
linewidth=2,
label="Random baseline"
)

ax.set_title("ROC Curve", fontsize=20, fontweight="bold", pad=16)
ax.set_xlabel("False Positive Rate", fontsize=13, labelpad=10)
ax.set_ylabel("True Positive Rate", fontsize=13, labelpad=10)
ax.set_xlim(0, 1)
ax.set_ylim(0, 1.02)
ax.grid(True, linestyle="--", alpha=0.22)
ax.legend(frameon=True, facecolor="white", edgecolor="#E5E7EB", fontsize=12, loc="lower right")

for spine in ax.spines.values():
spine.set_visible(False)

fig.text(
0.5, 0.98,
"Threshold performance on hold-out test set",
ha="center",
fontsize=12,
color="#6B7280"
)

plt.show()
```

Purpose: This cell is part of the end-to-end healthcare machine learning workflow. It contributes to data ingestion, validation, preprocessing, feature engineering, model development, evaluation, visualization, or deployment activities depending on its position in the notebook.

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Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.

Cell 26: Technical Explanation

Code Overview

```
sns.set_theme(style="whitegrid", context="talk")
plt.rcParams.update({
"figure.facecolor": "white",
"axes.facecolor": "#FCFCFC",
"axes.edgecolor": "#D0D0D0",
"axes.labelcolor": "#1f2937",
"text.color": "#1f2937",
"xtick.color": "#374151",
"ytick.color": "#374151",
"font.family": "DejaVu Sans"
})

model = best_pipe.named_steps["model"]

try:
ohe = best_pipe.named_steps["preprocess"].named_transformers_["cat"].named_steps["encoder"]
cat_features = list(ohe.get_feature_names_out(all_cat_cols))
feature_names = all_num_cols + cat_features

if hasattr(model, "feature_importances_"):
importances = pd.DataFrame({
"feature": feature_names[:len(model.feature_importances_)],
"importance": model.feature_importances_
}).sort_values("importance", ascending=False)

display(importances.head(15))

top = importances.head(15).sort_values("importance", ascending=True)

fig, ax = plt.subplots(figsize=(11, 8), constrained_layout=True)
colors = sns.color_palette("viridis", n_colors=len(top))

ax.barh(
top["feature"],
top["importance"],
color=colors,
edgecolor="white",
linewidth=1.0
)

ax.set_title(
f"Top Feature Importances - {best_name}",
fontsize=20,
fontweight="bold",
pad=16
)
ax.set_xlabel("Importance", fontsize=13, labelpad=10)
ax.set_ylabel("")
ax.grid(axis="x", linestyle="--", alpha=0.25)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.spines["left"].set_visible(False)
```

```
ax.spines["bottom"].set_color("#D0D0D0")
ax.tick_params(axis="y", labelsize=12)
ax.tick_params(axis="x", labelsize=11)

for i, v in enumerate(top["importance"]):
ax.text(
v + max(top["importance"]) * 0.01,
i,
f"{v:.3f}",
va="center",
fontsize=11,
fontweight="bold"
)
plt.show()
except Exception as e:
print("Feature importance not available:", e)
```

Purpose: This cell is part of the end-to-end healthcare machine learning workflow. It contributes to data ingestion, validation, preprocessing, feature engineering, model development, evaluation, visualization, or deployment activities depending on its position in the notebook.

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Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.

Cell 27: Technical Explanation

Code Overview

```
Path("output").mkdir(exist_ok=True)

bench_df.to_csv("output/cv_benchmark.csv", index=False)
tuning_df.to_csv("output/tuning_results.csv", index=False)
final_metrics_df.to_csv("output/final_metrics.csv")
df.to_csv("output/cleaned_healthcare_data.csv", index=False)

print("All outputs saved in output/")
```

Purpose: This cell is part of the end-to-end healthcare machine learning workflow. It contributes to data ingestion, validation, preprocessing, feature engineering, model development, evaluation, visualization, or deployment activities depending on its position in the notebook.

Why This Code Exists: Every component is included to improve model reliability, reproducibility, maintainability, and business usefulness. The workflow follows common enterprise data science practices including data quality checks, missing value handling, pipeline construction, cross-validation, model benchmarking, hyperparameter tuning, and artifact generation.

Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.

Cell 28: Technical Explanation

Code Overview

```
print("Project completed successfully.")
print("Best tuned model:", best_name)
print("Best CV accuracy:", round(best_score, 4))
print("Final test accuracy:", round(final_metrics["Accuracy"], 4))
```

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Why This Code Exists: Every component is included to improve model reliability, reproducibility, maintainability, and business usefulness. The workflow follows common enterprise data science practices including data quality checks, missing value handling, pipeline construction, cross-validation, model benchmarking, hyperparameter tuning, and artifact generation.

Business Value: The objective is to support disease-risk prediction and help transform raw patient information into actionable insights for healthcare decision support.